00183

Sri Lanka Institute of Information Technology

B.Sc. Honours Degree in Information Technology

Specialized in Computer Systems & Networking

Final Examination
Year 3, Semester 1 (2024)

IE3080 – Network Security Engineering

Duration: 2 Hours

November 2024

Instructions to Candidates:

- This paper has 04 questions.
- Answer all questions in the booklet given.
- The total marks for the paper is 100.
- This paper contains 04 pages, including the cover page.
- This paper proceeds with 10 minutes of reading time.
- Calculators are allowed but electronic devices capable of storing and retrieving text are not allowed.

Question 01 — 25 Marks

a. A hybrid cloud is a cloud computing environment that combines on-premises data centers (private clouds) with public clouds, allowing organizations to run applications and manage data seamlessly across these environments. It provides a flexible, cost-effective IT infrastructure by unifying different computing resources. Your organization is planning to transition to a cloud-based infrastructure while managing network security for a hybrid cloud environment. You are tasked with ensuring the security of this infrastructure, mitigating both internal and external threats. Design a comprehensive security plan to secure the organization's hybrid cloud infrastructure. Your plan should address the following.

i. Identify and explain the potential network security threats (both internal and external) that are relevant to this transition. (10 Marks)

ii. Propose security strategies to mitigate these threats, including technologies, processes, and policies that need to be implemented. (10 Marks)

iii. Evaluate the effectiveness of your proposed strategies by discussing possible challenges or limitations and how they could be addressed. (5 Marks)

Question 02 — 25 Marks

a. Your organization is planning to implement a Public Key Infrastructure (PKI) to ensure secure communication between internal systems, external partners, and customers. As the network security engineer, you are responsible for designing and deploying the PKI system.

i. Analyze the components of a PKI system and explain how they ensure secure communication. Discuss the roles of Certificate Authorities (CAs), digital certificates, and certificate revocation in maintaining security. (10 Marks)

ii. Design a PKI deployment strategy for a large-scale organization, illustrating how internal and external parties would securely exchange data using public key cryptography. Provide a detailed process flow that includes key generation, certificate issuance, and certificate management. (10 Marks)

iii. Evaluate the limitations of PKI in the context of scalability, certificate management, and interoperability with third-party systems. Propose solutions to mitigate these challenges. (5 Marks)

Question 03 — 25 Marks

a. A small-to-medium-sized enterprise (SME) with multiple branch offices across Sri Lanka needs to establish a secure and reliable network connection between its headquarters and branches. The company is considering using a VPN (Virtual Private Network) solution to achieve this. (8 marks)

i. Above specific requirements of the SME, which VPN algorithm(s) and connection technology would be most suitable for their network?

ii. State your analysis and recommendations for the above customer's current need relating to VPN Technologies.

b. Considering the above SME's increasing complexity and scale of their networks, what are the key advantages and disadvantages of using point-to-point VPNs for their network infrastructures? (8 marks)

c. In industry, the complexity of network environments, and the growing importance of user authentication, what are the key advantages and disadvantages of using AAA (Authentication, Authorization, and Accounting) solutions, and how can they be effectively implemented to enhance network security and user experience? (9 marks)

Question 04 — 25 Marks

a. Given the increasing sophistication of cyber threats, how can a Cisco ASA be effectively configured to provide robust perimeter security for a large enterprise network?

   i. Please ensure to evaluate your answer using, Firewall Rules, Intrusion Prevention Systems (IPS), VPN Tunnels, Content Filtering, Advance Threat Protection (ATP), High Availability, Monitoring, and Logging. (12 marks)

b. In a large University network, how can a DMZ (Demilitarized Zone) be effectively implemented to provide robust security and protect sensitive data while ensuring seamless connectivity for students, faculty, and staff? Discuss and evaluate the DMZ functions and their benefits. (8 marks)

c. Considering the evolving threat landscape and the intricate nature of contemporary networks, what are the key strategies and best practices for leveraging Cisco ASDM (Adaptive Security Device Manager) to enhance network security, streamline management, and optimize performance in large-scale enterprise environments? (5 marks)

~ End of question paper ~